

EC-467 Mobile Application Development



BS (CE)-2021 LAB REPORT #11

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

Dr. Muhammad Bilal

LAB REPORT #11

Android Services

Introduction:

A Service is an application component that can perform long-running operations in the background. It does not provide a user interface. Once started, a service might continue running for some time, even after the user switches to another application. Additionally, a component can bind to a service to interact with it.

For example, a service can handle network transactions, play music, perform file I/O, or interact with a content provider, all from the background.

Tools:

Android Studio Jellyfish

Objectives of the experiment:

- i.To get basic understanding regarding android services.
- ii.To get basic understanding regarding how to start and stop android services.
- iii.To get to know about the benefits related to android services.

Class Task

JAVA CODE OF Main Activity:

```
package com.huzaifashahbaz.androidserviceapp;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
public class MainActivityService extends AppCompatActivity {
    Intent serviceIntent;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void startService(View view){
        serviceIntent = new Intent(getApplicationContext(), MyService.class);
        startService(serviceIntent);
    }
    public void stopService(View view){
        serviceIntent = new Intent(getApplicationContext(), MyService.class);
        stopService(serviceIntent);
    }
}

```

JAVA CODE OF My Service:

```

package com.huzaihashahbaz.androidserviceapp;
import android.app.Service;
import android.content.Intent;
import android.os.IBinder;
import android.util.Log;
import androidx.annotation.Nullable;
public class MyService extends Service {
    @Nullable
    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        serviceThread thread = new serviceThread(startId);
        thread.start();
        return START_STICKY;
    }
    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d("MyService", "onStartCommand: Service Stopped");
    }
    class serviceThread extends Thread {
        int threadID;
        serviceThread(int threadID) {

```

```

        this.threadID = threadID;
    }
    @Override
    public void run() {
        try {
            for (int i = 0; i < 10; i++) {
                Log.d("MyService", "serviceRunnable Running i = " + i);
                Thread.sleep(1000);
            }
        } catch (Exception e) {
            Log.d("MyService", "serviceRunnable Error " + e);
        }
        stopSelf(threadID);
    }
}

```

XML CODE OF Main Activity:

```

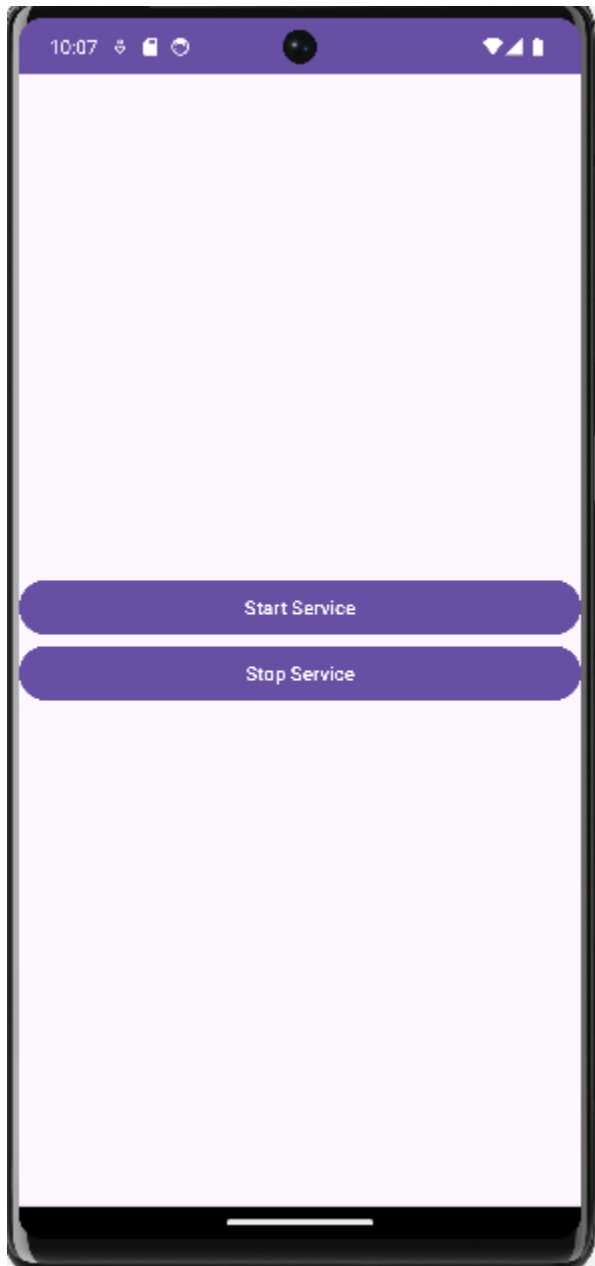
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <Button
        android:id="@+id/start_service_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="startService"
        android:text="Start Service"/>
    <Button
        android:id="@+id/stop_service_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="stopService"
        android:text="Stop Service"/>
</LinearLayout>

```

Results:

```
Run: app
2024-06-06 10:06:05: Launching app on 'Pixel 6 Pro API 34'.
Starting: Intent { act=android.intent.action.MAIN cat=[android.intent.category.LAUNCHER] cmp=com.huzaifashahbaz.androidserviceapp/.MainActivityService }
Open logcat panel for emulator Pixel 6 Pro API 34
```

OUTPUT:



Conclusion:

In conclusion, I had learn how to make android services by

using Java and xml. And also, I checked its function and working of android services in mobile app.